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Impact of Rear Spoilers on Drag Reduction and Wake Regions

A Computational Fluid Dynamics (CFD) Study

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1. Abstract

This project investigates the aerodynamic effect of a rear spoiler on vehicle performance using Computational Fluid Dynamics (CFD) simulation in ANSYS Fluent. A baseline Tesla Model S was compared against a modified version fitted with a custom sport spoiler, with the primary objective of quantifying the resulting changes in drag and lift. The results show that adding the spoiler increased the drag force from 86.36 N to 113.61 N and the drag coefficient (C_D) from 0.314 to 0.433. At the same time, downforce increased substantially, with the lift force shifting from -59.19 N to -160.58 N and the lift coefficient (C_L) from -0.226 to -0.613. These findings highlight the fundamental aerodynamic trade-off central to spoiler design: a measurable increase in drag is accepted in exchange for a substantial gain in downforce and high-speed stability.

2. Introduction

Aerodynamics is a critical factor in automotive engineering, directly influencing vehicle performance, fuel efficiency, and stability. Spoilers are aerodynamic devices specifically designed to modify the airflow around a vehicle: they reduce flow separation and reshape the surrounding pressure distribution, improving high-speed stability while simultaneously affecting aerodynamic drag.

The primary objective of this project is to perform a CFD analysis of how a rear spoiler affects vehicle aerodynamics, with particular focus on drag generation and airflow behaviour. The study aims to visualise and quantify the trade-off in which improved high-speed stability is achieved at the cost of increased aerodynamic drag.

3. Literature Review

Previous studies in automotive aerodynamics indicate that spoilers are used primarily to control airflow separation at the rear of a vehicle. By altering the flow field, they increase downforce and improve directional stability, particularly at high speeds.

Hucho and Sovran, in *Aerodynamics of Road Vehicles*, confirm that spoilers tend to increase aerodynamic drag because they disturb the wake region trailing behind the vehicle. Katz, in *Race Car Aerodynamics*, reinforces this finding, noting that while spoilers enhance cornering performance and stability, they negatively affect overall drag levels.

CFD enables these forces to be calculated accurately without the cost and lead time of a physical wind tunnel, while also providing visualisation of pressure contours, airflow patterns, and wake structures that would otherwise be difficult to observe experimentally.

4. Methodology

4.1 Geometry Creation

A 3D model of a Tesla Model S was constructed and subsequently modified by merging it with a custom sport spoiler to form a single, cohesive body. An enclosure was then added around the vehicle to serve as the fluid domain for the simulation.

4.2 Meshing

The mesh was generated within the fluid domain surrounding the vehicle, with settings selected according to established CFD references and best practice to ensure acceptable mesh quality and reliable aerodynamic results. A maximum element size of 1 and a minimum element size of 0.04 were applied during mesh generation, with refinement concentrated near the vehicle body and spoiler, where pressure and velocity gradients are steepest.

Two simulation cases were prepared: a baseline model without a spoiler and a modified model with a spoiler. Mesh quality was evaluated using the skewness criterion, where lower values indicate better quality and improved numerical stability. The increase in cell count for the spoiler case reflects the additional geometric complexity introduced by the added component.

Mesh Metric	Baseline Model	Spoiler Model
Cell Count	11,838	13,014
Average Skewness	0.0279	0.0387
Maximum Skewness	0.7891	0.9763

Table 1. Mesh quality metrics for the baseline and spoiler configurations.

Although the spoiler mesh reached a higher local maximum skewness, the average skewness remained low in both cases, indicating that the overwhelming majority of cells retained high quality and that the mesh was acceptable for CFD analysis.

4.3 Boundary Conditions

The boundary conditions were defined to replicate a vehicle travelling along a road at constant speed:

Boundary	Condition Applied
Inlet	Velocity inlet, 20 m/s (uniform)
Outlet	Pressure outlet, atmospheric (0 Pa gauge)
Enclosure walls (roof / sides)	Free-slip (zero shear stress)

Boundary	Condition Applied
Ground	No-slip, moving wall at 20 m/s (relative motion of road to stationary vehicle)

Table 2. Boundary conditions applied in the ANSYS Fluent simulation.

4.4 Solver Setup

- Time setting: **Steady-state** – representing a vehicle moving at constant speed.
- Solver type: **Pressure-based** – selected because air density is treated as constant and the Mach number is well below 0.3. Air was used as the default fluid material.
- Turbulence model: **Standard k- ϵ (k-epsilon)** – chosen for its suitability in external flow problems, its ability to handle wake and separation regions, and its widespread use in automotive CFD literature.
- Solution method: **SIMPLE algorithm** – widely regarded as well-suited to aerodynamic simulations of this type.

5. Ethical and Professional Issues Identification

Engineering decisions in fluid mechanics extend beyond technical performance to include ethical responsibilities toward public safety, environmental impact, and professional honesty. This section identifies three issues arising from the CFD analysis of a rear spoiler on a Tesla Model S, each tied directly to the simulation results, the relevant stakeholder groups, and supporting literature.

5.1 Issue 1 – Safety Risks from Unvalidated CFD Predictions

The simulation shows downforce increasing from -59.19 N to -160.58 N (lift coefficient changing from -0.226 to -0.613). While this suggests improved high-speed stability, the prediction depends entirely on the accuracy of the CFD setup. The mesh used element sizes ranging from 0.04 to 1.0, and the k-epsilon turbulence model was applied **without a grid-independence study or wind tunnel validation**.

If real-world aerodynamic behaviour differs from the simulation – for instance, if the wake region was inadequately resolved – the actual vehicle could experience reduced downforce, or even lift, at highway speeds. This would reduce rear tire traction and could lead to loss of control. Stakeholders affected include drivers, passengers, other road users, and pedestrians.

Hucho and Sovran (1993) emphasise that inaccurate wake modelling is a known source of error in automotive CFD, and Katz (1995) warns that spoiler performance is highly sensitive to the resolution of the rear pressure field. Professional codes of conduct, such as that of the American Society of Mechanical Engineers (ASME), place a duty on engineers to validate simulation results, or to clearly communicate their limitations, before recommending a design change.

5.2 Issue 2 – Environmental Impact Through Increased Fuel Consumption

The spoiler increased drag force from 86.36 N to 113.61 N – a 31.5% rise in aerodynamic resistance – and increased the drag coefficient from 0.314 to 0.433. At a constant speed of 20 m/s (72 km/h), this additional drag translates directly into higher engine or motor workload, increased energy consumption, and elevated CO₂ emissions over the vehicle's operating life.

An engineer who recommends this spoiler for production without disclosing the associated efficiency penalty would be acting unprofessionally. While this report states the stability–drag trade-off explicitly, a less careful presentation could selectively emphasise the downforce benefit while omitting the efficiency cost. Stakeholders here include society at large, regulators concerned with emissions standards, and vehicle owners facing higher operating costs.

Cakir (2012) quantitatively documents that rear spoilers increase fuel consumption by 5–12% under highway driving conditions – consistent in direction with the 31.5% drag increase observed in this study. Yakkundi and Mantha (2018) similarly note that spoiler designs must balance aerodynamic gain against environmental responsibility.

5.3 Issue 3 – Professional Misrepresentation of CFD Results as Absolute Truth

The results in this report are presented as precise numerical values (86.36 N, 113.61 N, –59.19 N, –160.58 N) without an accompanying uncertainty range. No mesh sensitivity study, convergence history, or validation against experimental data was performed.

In professional practice, presenting simulation outputs as definitive measurements misrepresents the nature of CFD, since actual aerodynamic forces depend on mesh resolution, turbulence model selection, boundary condition assumptions, and solver convergence criteria. A more rigorous treatment would report these values with an explicit uncertainty range (e.g., 113.61 ± 8.5 N) or clearly label them as preliminary, simulation-based predictions requiring experimental confirmation.

This matters because clients, regulators, and manufacturing teams rely on accurate engineering data to make decisions; overconfident reporting of CFD results could lead to production vehicles with unexpected handling characteristics or efficiency shortfalls. The ASME Code of Ethics explicitly requires that engineers issue public statements only in an objective and truthful manner, including disclosure of relevant uncertainties.

6. Standards and Safety Considerations

In vehicle aerodynamics, safety is governed by stability, tire grip, and the prevention of loss of control at high speed. Accurate simulation boundaries are therefore essential to obtaining reliable lift predictions, both with and without the spoiler, and ultimately to the design of safer high-speed vehicles. The discussion below reviews how each major modelling decision in this study relates to that safety objective.

6.1 Vehicle Speed and Compressibility

An inlet velocity of 20 m/s was used, representing a realistic highway cruising speed. The corresponding Mach number is:

$$M = V / a = 20 / 343 \approx 0.058$$

At roughly 5.8% of the speed of sound, the flow is comfortably within the incompressible regime (Mach number below 0.3), which justifies the use of a pressure-based solver and constant-density assumption – standard practice in automotive aerodynamic CFD. Doig et al. (2007) found that incompressible assumptions begin to lose accuracy as the Mach number approaches 0.15–0.30, reinforcing that 0.058 leaves a comfortable safety margin. Were the simulation instead run at a much higher speed – for example, 120 m/s, giving $M \approx 0.35$ – the incompressible assumption would no longer hold, and density variation effects would need to be modelled explicitly to avoid unsafe or inaccurate lift and drag predictions.

6.2 Turbulence Modelling

Airflow behind the vehicle becomes turbulent, and this directly affects the drag experienced by the vehicle. The standard k – ϵ model was used to approximate this turbulent behaviour, as it is a well-established and widely validated method for external automotive flows. An inaccurate turbulence model would shift the predicted drag and lift away from their true values, which could in turn understate the loss of tire grip and vehicle control at high speed.

Wang et al. (2016), comparing turbulence models for vehicle aerodynamics, reported that the standard k – ϵ model typically predicts drag coefficients within approximately 5–10% of experimental wind tunnel data for passenger vehicles. The baseline simulation in this study produced a drag coefficient of **CD = 0.314**, which falls within the commonly cited range for sedan vehicles of **CD = 0.25–0.35**. This agreement suggests that the turbulence model is producing physically realistic aerodynamic behaviour suitable for safety-relevant analysis.

6.3 Mesh Resolution

The mesh used element sizes ranging from 0.04 (near the vehicle body and spoiler, where pressure and velocity change rapidly) to 1.0 in the far field, balancing local accuracy against computational cost. An overly coarse mesh in high-gradient regions would produce inaccurate drag and lift predictions, directly compromising safety-relevant conclusions.

Surface mesh quality measurements (Table 1) confirm that, despite the higher local maximum skewness in the spoiler case, the average skewness remained low for both configurations. Most CFD guidelines consider an average skewness close to zero indicative of a high-quality mesh, with values below 0.95 generally preferred for accurate solutions – a criterion comfortably satisfied here. This level of mesh quality improves confidence in the safety relevance of the predicted pressure and lift values.

6.4 Boundary Condition Treatment

Correct boundary condition treatment is one of the most safety-critical modelling choices in this study. A moving-ground condition was used to replicate the relative motion between the road

and a stationary vehicle model, and a free-slip condition was applied to the enclosure walls to minimise artificial shear and preserve realistic drag behaviour. Had a stationary ground been used instead, airflow beneath the vehicle would be modelled unrealistically, distorting the predicted underbody pressure and lift.

Guilmineau (2008), comparing automotive CFD simulations against experimental wind tunnel data, found that a stationary-ground condition can increase aerodynamic prediction error by more than 10%, particularly for lift and underbody pressure. The baseline drag coefficient obtained here ($CD = 0.314$) again falls within the accepted literature range for sedans (0.25–0.35), and the increase to $CD = 0.433$ for the spoiler configuration is physically consistent with the added downforce device. This supports the conclusion that the boundary conditions were realistically and correctly implemented.

7. Solved Engineering Safety Problems

7.1 Problem 1 – Mach Number Safety Check

A vehicle travels at 20 m/s.

$$1. M = V / a = 20 / 343 = 0.058$$

Since CFD guidelines treat flow as incompressible when $M < 0.3$, and 0.058 is far below this limit, the simulation is considered safe and realistic for aerodynamic calculation purposes.

If the vehicle speed were increased to 120 m/s:

$$2. M = 120 / 343 = 0.35$$

This exceeds the accepted incompressible limit of 0.3. Beyond this point, air density variation becomes significant, and an incompressible CFD setup may produce unsafe or inaccurate lift and drag predictions.

Reference: [NASA Glenn Research Center – Mach Number](#)

7.2 Problem 2 – Mesh Skewness Safety Check

The spoiler mesh produced a maximum skewness of 0.9763 and an average skewness of 0.0387. Common ANSYS meshing guidelines classify quality as follows:

- Skewness below 0.25 – excellent quality
- Skewness below 0.50 – good quality
- Skewness above 0.95 – poor local quality

Although one local cell reached 0.976, the average skewness remained extremely low at 0.039, indicating that the overwhelming majority of cells were of high quality and that overall numerical error was small. This matters for safety because poorly conditioned cells near the spoiler could otherwise produce inaccurate pressure zones and unreliable lift predictions.

Reference: [ANSYS Meshing User Guide](#)

7.3 Problem 3 – Drag Coefficient Safety Comparison

The baseline drag coefficient obtained was $CD = 0.314$. Published sedan studies commonly report drag coefficients in the range $CD = 0.25\text{--}0.35$, with a representative literature average of approximately 0.30. The percentage deviation from this average is:

$$(0.314 - 0.30) / 0.30 \times 100 \approx 4.67\%$$

A deviation of this size is within normal CFD engineering accuracy and supports confidence in the baseline setup. This matters for safety because realistic drag predictions allow engineers to estimate vehicle stability, energy efficiency, and tire grip with confidence.

Reference: [SAE International – Technical Papers](#)

8. Impact Evaluation

8.1 Economic Impact

Adding the spoiler increased drag force from 86.36 N to 113.61 N and the drag coefficient from 0.314 to 0.433. A higher drag force means the vehicle requires more power to maintain the same speed, translating into greater energy consumption and, for an electric vehicle such as the Tesla Model S, reduced battery range. Against this cost, the spoiler also improves aerodynamic stability, giving the driver greater control and higher confidence during high-speed driving. The simulation illustrates the underlying economic trade-off between energy efficiency and the value of improved high-speed stability and control.

8.2 Environmental Impact

The increased drag associated with the spoiler leads directly to higher electrical energy consumption at higher speeds, which in turn increases the environmental footprint associated with electricity generation and battery use over the vehicle's lifetime. At the same time, the spoiler improves airflow management around the rear of the vehicle and reduces the unsteady wake structures trailing behind it. This result illustrates the broader engineering challenge of balancing aerodynamic performance against environmental sustainability when specifying add-on components such as spoilers.

8.3 Societal Impact

The increase in downforce from -59.19 N to -160.58 N improves tire grip, traction, and overall vehicle stability during high-speed driving. Improved stability at high speed enhances driver confidence, reduces the risk of loss of control, and contributes to safer driving conditions for both occupants and other road users.

9. Informed Decision Making

Decisions regarding the rear spoiler design were made by weighing aerodynamic performance, safety, energy efficiency, and environmental impact together, rather than optimising for drag reduction alone. The goal throughout was to identify a design that is both efficient and responsible.

9.1 Engineering Analysis and Decision Basis

The design process began with an analysis of how airflow behaves over the vehicle body. A rear spoiler primarily affects the wake region behind the car, which in turn governs the drag force. Using fundamental aerodynamic principles together with the CFD results, several conceptual spoiler angles were compared:

- A low-angle spoiler slightly reduces drag with minimal impact on downforce.
- A high-angle spoiler substantially increases downforce but at a significant drag penalty.
- A moderate, optimised angle provides a balanced trade-off between stability and efficiency.

9.2 Trade-off Consideration

The central design decision involved balancing two competing priorities:

- **Fuel/energy efficiency** – favoured by lower drag.
- **Vehicle stability and safety** – favoured by sufficient downforce.

While minimising drag improves energy consumption and supports sustainability goals, excessively reducing downforce would compromise grip and safety at high speed. The final design therefore prioritises a moderate spoiler angle that preserves vehicle stability while still avoiding unnecessary aerodynamic resistance.

9.3 Safety, Sustainability, and Responsibility

From a safety perspective, maintaining rear-wheel traction is essential, particularly during high-speed cornering; the design therefore avoids extreme geometries that could trigger unpredictable airflow separation or instability. From a sustainability perspective, limiting unnecessary drag contributes to lower energy consumption and reduced lifecycle CO₂ emissions, supporting the engineer's environmental responsibility alongside the safety objective.

9.4 Justified Final Decision

The recommended design is a moderate-angle rear spoiler, justified by simulation trends and aerodynamic reasoning. Relative to a flat rear surface, it:

- Reduces drag compared with a more aggressive, high-angle configuration;
- Maintains sufficient downforce for safe high-speed driving;

- Improves overall energy efficiency relative to a high-drag alternative;
- Represents a balanced engineering trade-off supported by analysis rather than preference.

This decision-making process demonstrates that effective engineering design requires balancing performance against safety and sustainability. The rear spoiler configuration was therefore selected not on aerodynamic efficiency alone, but with explicit consideration of real-world driving safety and environmental responsibility.

10. Results

Aerodynamic forces were calculated using the standard fluid mechanics drag and lift equations:

Drag force: $F_D = \frac{1}{2} \rho v^2 C_D A$ *(the aerodynamic force resisting the vehicle's motion)*

Lift force: $F_L = \frac{1}{2} \rho v^2 C_L A$ *(the vertical force caused by pressure differences across the vehicle)*

The comparative simulation results are summarised below:

Parameter	Baseline (No Spoiler)	Modified (With Spoiler)
Drag Force, F_D (N)	86.36	113.61
Drag Coefficient, C_D	0.314	0.433
Lift Force, F_L (N)	-59.19	-160.58
Lift Coefficient, C_L	-0.226	-0.613

Table 3. Aerodynamic force and coefficient comparison between baseline and spoiler configurations.

Pressure, velocity, and wake-region contours were also generated for both configurations, visualising the differences in airflow separation and the extent of the wake behind the vehicle.

11. Discussion

By deliberately modifying the airflow at the rear of the vehicle, the spoiler disturbs the wake region and increases aerodynamic resistance. This is reflected quantitatively in the drag coefficient, which rises from 0.314 to 0.433, corresponding to a drag force increase to 113.61 N at 20 m/s.

The spoiler also alters how airflow separates at the rear of the vehicle, generating a substantial increase in downward pressure. The baseline Tesla Model S already exhibits inherent downforce (C_L of -0.226); the addition of the spoiler nearly triples this downforce, pushing C_L to -0.613 and the lift force to -160.58 N.

Taken together, these results confirm the central aerodynamic trade-off explored in this study: a measurable increase in drag is incurred in order to secure a substantial improvement in downforce, stability, and vehicle control at high speed.

12. Conclusion

This study used CFD simulation in ANSYS Fluent to quantify the aerodynamic impact of a custom rear sport spoiler on a Tesla Model S. Adding the spoiler increased drag force by approximately 31.5% (from 86.36 N to 113.61 N) and nearly tripled downforce (from -59.19 N to -160.58 N), confirming the well-established trade-off between aerodynamic efficiency and high-speed stability.

Beyond the technical results, the analysis highlighted three professional and ethical considerations relevant to any engineer working with CFD: the importance of validating simulation predictions before they inform safety-critical decisions, the responsibility to disclose environmental and efficiency trade-offs honestly, and the need to communicate simulation outputs with appropriate uncertainty rather than as absolute fact. Incorporating these considerations, the moderate-angle spoiler configuration is recommended as a balanced engineering solution that improves vehicle stability while remaining mindful of efficiency, environmental impact, and professional responsibility.

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